# 🛡️ OmniSight: Official Hackathon Project Report

**Submission for:** SerpApi India Hackathon 2026
**Project Name:** OmniSight - Autonomous Enterprise Intelligence
**Track/Focus:** AI Agents, Market Intelligence, Data Aggregation

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## 1. Executive Summary

Modern market research is a fragmented, tedious process. Entrepreneurs, e-commerce dropshippers, and marketing agencies spend countless hours switching between trend analysis tools, keyword planners, retail pricing engines, and news aggregators to validate a single business hypothesis.

**OmniSight** is an agentic AI command center designed to eliminate this friction entirely. By leveraging the immense concurrent power of **SerpApi** and the generative intelligence of **Google Gemini**, OmniSight transforms a simple keyword input into a comprehensive, multi-phase Startup Blueprint and visual Threat Web in a matter of seconds.

## 2. The Problem Statement

- **Data Fragmentation:** Crucial market signals (Historical trends, retail ceilings, SEO gaps, brand safety) are scattered across the web.
- **Analysis Paralysis:** Raw data is useless without strategic interpretation.
- **High Barrier to Entry:** Elite enterprise intelligence tools (like Bloomberg Terminals or Enterprise SEO suites) cost tens of thousands of dollars annually, gating indie founders and small agencies out of predictive data.

## 3. The OmniSight Solution

OmniSight is a Next.js 14 Web Application that acts as a single pane of glass for market entry strategy. Instead of providing the user with raw data tables,

OmniSight interprets the data using LLMs and visualizes it using interactive node-graphs ( `ReactFlow` ) and predictive radar charts ( `Recharts` ).

**Key Differentiators:**

1. **Agentic Autonomy:** The user does not need to analyze the data. The AI acts as a dedicated Market Strategist, writing out a 12-month execution phase plan based entirely on the live SerpApi metrics.
2. **Interactive Live Chat:** Users aren't just reading static reports; they can actively converse with the AI agent in the sidebar, interrogating the market data to refine their business strategies.
3. **Beautiful, Dark-Mode SaaS UI:** OmniSight steps away from the "hackathon toy" aesthetic and utilizes a premium, professional interface designed for high-value enterprise users.
4. **Massive Concurrency:** We don't just query Google once. We hit 5 distinct Google ecosystems simultaneously to paint a holistic picture of the market.

# 4. Technical Architecture & Integrations

## Frontend Stack

- **Framework:** Next.js 14 (App Router)
- **Styling:** Tailwind CSS (Dark theme with glowing neon accents)
- **Visualizations:** `Recharts` for Market Threat Radars, `ReactFlow` for Interactive SEO consumer intent mapping.
- **Icons:** Lucide React

## Backend Orchestration (SerpApi)

The core engine runs on a custom Node.js `/api/analyze` route. When a user submits a query, OmniSight uses `Promise.all` to concurrently trigger five distinct SerpApi engines:

1. **`google_trends`** : Fetches 10-year historical macro-viability data.
2. **`google_shopping`** : Scrapes high/low pricing arrays to establish retail ceilings and wholesale arbitrage floors.

3. `google` **(Organic)**: Extracts "Related Searches" and "People Also Ask" arrays to discover long-tail gaps.
4. `google_news` : Pulls recent headlines for brand safety and sentiment analysis.
5. `youtube` : Scrapes video IDs to natively embed high-performing media reviews into the dashboard.

### Generative AI Pipeline (Google Gemini)

Once the SerpApi payload is successfully aggregated, the raw JSON is compressed and passed to our serverless endpoints ( `/api/blueprint` and `/api/chat` ).

- **Dynamic Blueprints:** We built a hyper-specific mega-prompt instructing Gemini 1.5 Flash to output a deeply nested JSON object. The frontend safely parses this JSON and maps it directly into the UI components (Executive Summary, Market Gap Analysis, and Visual Roadmap).
- **Interactive Conversational AI:** The `/api/chat` route processes the user's live queries against the active SerpApi context, allowing them to ask follow-up questions about the market data in real-time.
- *Fallback Mechanism:* OmniSight implements a robust `try/catch` fallback. If an API key is invalid or rate-limited during judging, the system gracefully catches the error and serves a simulated intelligence payload so the UI never crashes and the chat remains functional.

## 5. Business Viability & Market Opportunity

**Target Audience:**

- **Dropshippers & E-commerce:** Identifying arbitrage gaps between Alibaba/wholesale floors and Google Shopping ceilings.
- **SaaS Founders:** Validating software niches based on highly-specific "People Also Ask" technical queries.
- **Marketing Agencies:** Pitching clients with instantly generated, data-backed 12-month SEO roadmaps.

**Monetization Strategy:**

- **Freemium Tier:** 3 free searches/month to demonstrate the platform's power.
- **Pro Tier ($99/mo):** Unlimited searches, PDF export capabilities, and deeper historical trend lookbacks.
- **Enterprise Tier:** Custom API access for hedge funds requiring programmatic threat web generation.

## 6. Future Roadmap

1. **Automated Execution:** Post-analysis, OmniSight will feature a "Deploy" button that automatically provisions a Shopify MVP store or a Next.js landing page customized to target the exact SEO gap identified by the AI.
2. **Global Supplier Matching:** Integrating the Alibaba/1688 API to dynamically link the user to wholesale manufacturers for the searched product.
3. **Social Sentiment Layer:** Expanding the SerpApi integration to scrape TikTok and X (Twitter) for viral velocity tracking.

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*Developed with ❤️ for the SerpApi Hackathon.*